

Protein Structure Prediction by Evolutionary Multi-objective Optimization: Search Space Reduction by Using Rotamers

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Abstract. The protein structure prediction (PSP) problem is considered an open problem as there is no recognized "best" procedure to find solutions. Moreover, this problem presents a vast search space and the analysis of each protein conformation requires significant amount of computing time. We propose a reduction of the search space by using the dependent rotamer library. Also this work introduces new heuristics to improve the multi-objective optimization approach to this problem.

1 Introduction

Proteins have important biological functions such as the enzymatic activity of the cell, attacking diseases, transport and biological signal transduction, among others. They are chains of amino acids selected from a set of twenty elements. Whenever an amino acid chain is synthesized, it folds together and uniquely determines its 3D structure. Moreover, although the amino acid sequence of a protein provides interesting information, the functionality of a protein is exclusively determined by its 3D structure [1]. Thus, there is a high interest in knowing the 3D structure of any given protein. For example, as some proteins attack diseases, taking into account that protein functionality depends on their structure, it would be important to know this 3D structure to learn its correlation with the protein functionality. Thus, the protein 3D structure can aid in the design of efficient drugs. In this way, many useful applications involve determining the protein structure, and many disciplines, such as Medicine, Biochemistry, Biology and Engineering can take advantage of solutions to this problem.

It is possible to reach the 3D structure of a protein experimentally by using methods such as X-ray crystallographic and nuclear magnetic resonance (NMR). Nevertheless, these processes are quite complex and costly as they would require months of expert work and laboratory resources. This situation comes clear if considering that less than a 25% of the protein structures included in the PDB (Protein Data Bank) have been solved.

An alternative approach is to use high performance computing. Nevertheless, the computational analysis of each conformation requires a significant time and

this is a Grand Challenge Problem that still remains unsolved [1]. This computer approach is called protein structure prediction (PSP) and implies predicting the tertiary structure of a protein given its primary structure.

The protein structure can be divided into four levels (Fig. 1). *The primary structure* is the sequence of amino acids that define the order of the sequence. *The secondary structure* is a set of contiguous amino acids joined by some hydrogen bonds. Then, the super-secondary structure is the combination of two secondary structures by a short connecting peptide. *The tertiary structure* is a three-dimensional structure of a single sequence of a protein. All force-field atoms take part in this conformation. Finally, *the quaternary structure* refers to a protein built by two or more sequences. This structure defines the relations between the different sequences of the protein. Thus, the PSP tries to determine how the primary structure translates into the tertiary structure.

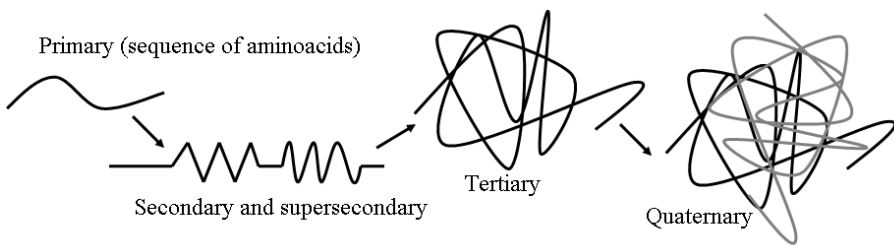


Fig. 1. Protein structures. From primary structure to quaternary structure.

The paper has been structured as follows. In Section II we introduce the concepts related with multi-objective optimization that we are going to use throughout of the work. The next section defines the different methods that our multi-objective optimization process needs in this problem. In the Section IV we provide the results and compare our algorithms with other approaches, and Section V exposes the conclusions.

2 PSP as an Optimization Process

Computational approaches to PSP can be divided into three main alternatives: comparative modeling, fold recognition and *ab initio* procedures [2, 3]. The first two alternatives consider that similar primary structures will fold into similar ways and try to take advantage of known structures for similar amino acid sequences (homologue sequences). The *ab initio* alternative does not require any homology and can be applied when an amino acid sequence does not correspond to any other known one. This paper uses the *ab initio* approach. In this case, to predict the protein 3D structure, we need to know the relation between primary and tertiary structures. Although this relation is not trivial, and there are many factors affecting the folding process that produces the final 3D conformation, we are looking for the native tertiary conformation with minimum free energy. Therefore, we do not take into account external factors that may influence the

protein folding process, such as temperature, neighbor proteins, and other conditions into the cell. We are considering Protein Structure Prediction (PSP) rather than Protein Folding (PF).

Many algorithms have been proposed to solve the PSP by optimizing an objective or energy function, usually by using evolutionary algorithms [4, 5]. Nevertheless, over the last few years, some new approaches have been suggested that model the PSP problem as a multi-objective problem [1, 6]. There are several reasons for trying a multi-objective approach. For example, as indicated in [6], there are works that demonstrate that some evolutionary algorithms improve their effectiveness when they are applied to multi-objective algorithms [7]. Indeed, in [1] it is argued that PSP problem can be naturally modelled as a multi-objective problem because the protein conformations could involve tradeoffs among different objectives as it is experimentally shown by analyzing the conflict between bonded and non-bonded energies. In this paper, we also apply a multi-objective approach to the PSP problem.

A multi-objective optimization problem [8] can be defined as the problem of finding a vector (1) that satisfies a given restriction set (2) and optimizes the function vector in (3). The objectives are usually in conflict between themselves, thus, optimizing one of them is carried out at the expense of the values of the others. This leads to the need of making a compromise, which implies the concept of Pareto optimality. In a multi-objective optimization problem, a decision vector x^* is said to be a Pareto optimal solution if there is not any other feasible decision vector, x , that improves one objective without worsening at least one of the other objectives, given P the set of Pareto optimal solutions, (4). Usually, there are many vectors which are Pareto optimal. These solutions are called non-dominated. The set of all non-dominated solutions, in the decision space, determines the Pareto front in the objective space. Finally, a decision maker can select the solutions in its preferred front zone [6].

$$x = [x_1, x_2, \dots, x_n] \quad (1)$$

$$g(x) \leq 0, h(x) = 0 \quad (2)$$

$$f(x) = \{f_1(x), f_2(x), \dots, f_m(x)\} \quad (3)$$

$$\forall a, b \in P(\exists i, j \in \{1, 2, \dots, n\} | (f_i(a) < f_i(b)) \wedge (f_j(a) > f_j(b))) \quad (4)$$

3 The Proposed Multi-objective Approach

This section describes the components required to solve the PSP problem by a multi-objective optimization using the *ab initio* approach. These components are: the cost function, initialization, the mutation operators and the set of variables.

Cost function: Although a realistic measure of protein conformation quality should probably imply considering quantum mechanics principles, it would be too computationally complex to become useful. Thus, as it is usual, we have used the Chemistry at HARvard Macromolecular Mechanics (CHARMM) and AMBER99 energy functions [1, 9]. These are the most popular all-atom force field used for studying macromolecules and proteins respectively. We have considered its implementation at the TINKER library package [10]. As we use a multi-objective evolutionary optimization formulation of the PSP problem, the different terms of the energy function have to be transformed into several objectives. In [1] it is distinguished between bond and non-bond energies. We use this idea, although we have introduced some modifications according to the characteristics of the solution domain. Analyzing these energies we can observe that the Van Der Wall energy term has higher change range than others. Accordingly, this last energy terms can be hidden by the Van Der Wall energy term. Thus to optimize this energy appropriately, we propose a cost function with three objectives as follows: the bond energy and two more objective for the non-bond energies, one for Van Der Wall and other for the rest of non-bond terms. Also, we propose to take in account the difference between the probabilistic 3D protein structure (see next paragraph) and the current protein structure. To do that, we use another cost function configuration: bond energy, non-bond energy and difference with the probabilistic protein 3D structure.

Initialization: We have developed two initialization methods: random and probabilistic. The random method sets each variable with a random value according to the constraints of the variable. The probabilistic method uses the rotamer libraries to set each amino-acid at its most probable conformation. In this work we do not use the probabilistic initialization in the NSGA2 algorithm, because a population algorithm requires a set of different individuals.

Mutation operators: In [1] there are two different mutation operators. The first one is used to explore the search space, and the second one performs a local search optimization. We propose a third mutation operator, this method generate a mutation more conservative than the mutation to explore, and bigger than the mutation to perform a local search. This operator is executed with the same probability as the first one, and it mutates the side-chain with a gaussian function defined in the rotamer library. Doing that, we can avoid some local minima.

Set of variables: Any cost function depends on a set of variables that define the search space. These variables provide the required information to build the 3D conformation of the protein. In this paper we use torsion angles to represent the conformation of the protein, because this representation needs less variables than other alternatives. Three torsion angles are required in the backbone per each amino acid and some additional torsion angles depending on the side-chain [1] (Figure 2).

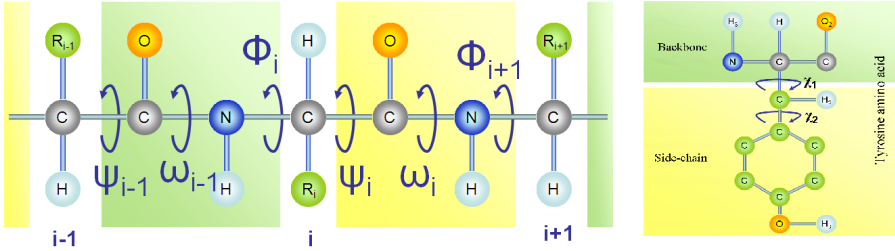


Fig. 2. (left) Backbone angles ϕ , ψ and ω . (right) Side-chain angles χ_i in the tyrosine amino acid.

In many optimization problems, once a suitable cost function is found, it would not be difficult to obtain an efficient evolutionary algorithm that generates sufficient good solutions for the problem. This is not the case in the PSP problem. As the search space is so vast, it is necessary to select an adequate set of variables and a representation space, along with the inclusion of specific information about the PSP problem in the evolutionary algorithm. Some alternatives can be taken into account to reduce the search space such as secondary and super-secondary structure prediction and inclusion of rotamers libraries [11]. Rotamers libraries have statistical information about the conformation of the amino acids in known molecules.

Thus, although the PSP problem implies to predict the tertiary structure of a given protein from its primary structure, it could be a good idea to use predictions of the secondary and super-secondary structures as they give us information about the amino acids involved in one of these structures, determining some constraints in the torsion angles of each amino acid (as shown in Table 1). In order to get the super-secondary structure given its secondary structure, we have to analyze the conformation of the residues in the short connecting peptide between two secondary structures. They are classified into five types, namely, a, b, e, l or t [12]. Sun *et al.* [12] developed a method to predict the eleven most frequently occurring super-secondary structures: H-b-H, H-t-H, H-bb-H, H-ll-E, E-aa-E, H-lbb-H, H-lba-E, E-aal-E, E-aaal-E and H-l-E where H and E are α helix and β strand, respectively. In this way a reduction in the search space of the PSP problem is obtained.

Moreover, side-chain torsion angles have interesting dependencies. Dumbrack *et al.* [11] give many rotamers libraries that help us to identify constraints about these torsion angles. An example of these libraries is the backbone-independent rotamer library. Given an amino acid, this library includes constraints for its side-chain torsion angles. There are dependencies between side-chain torsion angles in the same amino acid, but these dependencies are not related with the backbone torsion angles. It is difficult to find a good representation for these constraints because their mutual dependency. The backbone-dependent rotamer library is more complex than the backbone-independent rotamer library, because the former includes the dependency between side-chain torsion angles and backbone torsion angles. Therefore, the size of the library increases significantly and,

Table 1. Search space of each angle ϕ and ψ depending on the position of the super-secondary structure they are

Super-secondary structure	ϕ	ψ
H (α helix)	[-75, -55]	[-50, -30]
E (β strand)	[-130, -110]	[110, 130]
a	[-150, -30]	[-100, 50]
b	[-230, -30]	[100, 200]
e	[30, 130]	[130, 260]
l	[30, 150]	[-60, 90]
t	[-160, -50]	[50, 100]
undefined	[-180, 0]	[-180, 180]

although the information provided by this library is interesting, the time required to include this information in the optimization procedure also increases. Therefore, there are dependencies among backbone torsion angles and side-chain torsion angles, and there are also dependencies inside torsion angles of the side-chain. An optimization process tries an independent movement for each variable. Thus, if we include the backbone torsion angles and the side-chain torsion angles in the set of variables, we will ignore the cross-information between torsion angles. Therefore, a suitable way to approach the problem could be to implement a way to manage the backbone torsion angles in an optimization process and, inside it, to include another mechanism to manage the side-chain torsion angles.

Thus, In this work we propose a new method to manage torsion angles using the backbone-dependent rotamer library. In this way, we reduce the set of variables involved in the optimization process by eliminating the side-chain torsion angles. Nevertheless we can not eliminate side-chain torsion angles without adding another mechanism to take them into account. This mechanism selects the most probable conformation of the side-chain in each amino acid depending on the backbone torsion angles. Then it is possible change the side-chain conformation, considering the alternatives included in the library, to produce a better global conformation.

4 Results

In this section we provide and analyze the results obtained with our procedure. They have been obtained by using the 1CRN protein [1, 13] as a benchmark. As the cost functions used to model the PSP problem only approximate the conformation energy of the protein, it is not enough to measure the quality of the 3D structure obtained. To do so, we should accomplish a comparison between a given known protein structure and the solution obtained by optimizing the cost function that model the corresponding PSP problem. The most famous measure of similarity between predicted and known native structures is the RMSD [1] (5). RMSD computes the 3D difference between two structures, therefore the lower the molecule RMSD, the better is the 3D structure obtained.

$$RMSD(a,b) = \sqrt{\frac{\sum_{i=1}^n |\tau_{ai} - \tau_{bi}|^2}{n}}$$

(5)

We have executed an I-PAES and a NSGA2 with our heuristics and methods. We also have executed the original implementation of the algorithm described in [1]. We have executed each one for 21 times along 250.000 cost function evaluations and we have selected the best solution of the Pareto front of each execution. This information give us the best behavior that one approach has. We have computed the mean of these RMSDs obtaining the results of Table 2. The results shown in Table 2 do not allow us to get very clear conclusions about the best alternative for the nature of the third objective, the multi-objective evolutionary algorithm applied, and initialization procedure, or the energy functions (CHARMM or AMBER99). If we consider the obtained mean values of RMSD bellow 8.00 Å, it can be concluded that AMBER99 presents a better behavior and that the use of a third objective is useful. Nevertheless, although the best result is provided when NSGA2 is executed, I-PAES provides values below 8.00 Å in two alternatives. Similar things happen with the initialization and the third objective. This way, although the alternatives proposed in this paper allow an improvement of previous procedures in the multi-objective approach to PSP [1], it is clear that more experiments are required considering a wider set of proteins. This is the goal of or future work along with the parallel implementation of our procedures in order to reduce the times required to get the results. In [14], we provided our first results in this trend. In this work we have used 14 processors to execute parallel algorithms.

Table 2. Provided algorithms versus other approach for 1CRN protein

Algorithm	# Obj.	Init.	3rd obj.	mean	dev.	parallel	time
<i>NSGA2-Reduced-amber99</i>	2	Rand.	-	8.56 Å	0.99 Å	Yes	1.5 h
<i>NSGA2-Reduced-amber99</i>	3	Rand.	Van Der	7.71 Å	0.54 Å	Yes	1.5 h
<i>NSGA2-Reduced-amber99</i>	3	Rand.	Prob.	8.28 Å	0.96 Å	Yes	1.5 h
<i>I-PAES-Reduced-amber99</i>	2	Rand.	-	11.40 Å	1.78 Å	Yes	2.5 h
<i>I-PAES-Reduced-amber99</i>	2	Prob.	-	10.35 Å	1.50 Å	Yes	2.5 h
<i>I-PAES-Reduced-amber99</i>	3	Rand.	Prob.	9.72 Å	1.23 Å	Yes	2.5 h
<i>I-PAES-Reduced-amber99</i>	3	Prob.	Prob.	7.86 Å	1.03 Å	Yes	2.5 h
<i>I-PAES-Reduced-amber99</i>	3	Rand.	Van Der	8.49 Å	1.19 Å	Yes	2.5 h
<i>I-PAES-Reduced-amber99</i>	3	Prob.	Van Der	7.78 Å	0.94 Å	Yes	2.5 h
I-PAES CHARMM27 [1]	2	Rand.	-	8.35 Å	1.35 Å	No	20.0 h

5 Conclusion

The PSP problem joins biological and computational concepts. It requires accurate and tractable models of the conformations energy. Thus, there is a long way to go to find useful solutions to the problem for proteins of realistic sizes. Our contribution in this paper deals with a new procedure for PSP based on

a multi-objective evolutionary algorithm. It allows a reduction in the number of variables and few heuristics to improve the quality of the solutions such as the three objectives cost function. By comparing the results of our procedures with those provided by other multi-objective approach [1], it can be shown that our methods provide conformations of comparable or even better quality. Nevertheless, more experiments are required by using more (and more complex) proteins to get more conclusive results.

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